

Jiaming He

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EDUCATION

Oxford Brookes University

Software Engineering, *B.Eng.*

Sept. 2021 - July 2025

- British Diploma: Upper Second-class Honours (2:1)

RESEARCH INTERESTS

My research interest lies in **Trustworthy AI**. I focus on building robust and reliable methods to implement deep learning system in physical world. I have worked extensively on adversarial attacks, backdoor attacks, and data poisoning (unlearnable examples) as well as their mitigations. Recently, my interests extend to data privacy (e.g., protecting data from unauthorized commercial training and model inversion risks) and the security of foundation models such as LLMs and multimodal systems.

RESEARCH EXPERIENCE

Ant Group, Ant Research (CP Lab), Research Intern

May 2025 - Sept. 2025

During my internship at Ant Research (CP Lab), I conducted research on video privacy extraction with large video language models under the supervision of Dr. Cheng Hong, Dr. Zhicong Huang and Prof. Neil Gong (Duke University, Assoc. Professor). This work has been submitted to USENIX Security 2026.

UESTC, Research Assistant

May 2023 - Sept. 2025

Working at the Internet Data Security Laboratory under the supervision of Prof. Hongwei Li (**IEEE Fellow**), my research focuses on Artificial Intelligence Security. I primarily investigate the security of large-scale foundation models (e.g., jailbreaking Large Language Models and backdoor attacks in diffusion-based Text-to-Image scenarios). During this period, my work as first author has been accepted by CCF-A/B/C conferences including CVPR 2026, AAAI 2025 (Oral), ICME 2025 (Oral), ICASSP 2025, and ICC 2024 (Oral).

King Abdullah University of Science and Technology (KAUST), Research Intern

May 2024 - Nov. 2024

During my remote internship at KAUST, I conducted research on Machine Unlearning and Membership Inference Attacks (MIA), advised by Prof. Di Wang. I explored these vulnerabilities within mainstream vision-language multimodal models, such as CLIP and Llava.

PUBLICATIONS AND PRE-PRINTS

TEAR: Temporal-aware Automated Red-teaming for Text-to-Video Models,” IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (CCF-A, Top conference in Computer Vision)

Jiaming He, Guanyu Hou, Hongwei Li, Zhicong Huang, Kangjie Chen, Yi Yu, Wenbo Jiang, Guowen Xu, Tianwei Zhang

Accepted Feb. 2026

”Watch Out for Your Guidance on Generation! Exploring Conditional Backdoor Attacks against Large Language Models,” AAAI Conference on Artificial Intelligence (CCF-A, Top conference in Machine Learning) Oral (Top 5%)

Jiaming He, Wenbo Jiang, Hongwei Li, Guanyu Hou, Wenshu Fan, Rui Zhang

Accepted Nov. 2024

”Weaponizing Tokens: Backdooring Text-to-Image Generation via Token Remapping,” IEEE International Conference on Multimedia & Expo (ICME) (CCF-B, Top conference in Multimedia)

Jiaming He, Wenbo Jiang, Guanyu Hou, Qiyang Song, Hongwei Li

Accepted Mar. 2025

”PRESS: Defending Privacy in Retrieval-Augmented Generation via Embedding Space Shifting,” IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP) (CCF-B, Top conference in Signal Processing)

Jiaming He, Cheng Liu, Guanyu Hou, Wenbo Jiang, Jiacheng Li

Accepted Dec. 2024

”UTOPIA: Unlearnable Tabular Data via Decoupled Shortcut Embedding,” International Conference on Machine Learning (ICML) (CCF-A, Top conference in Machine Learning)

Jiaming He, Fuming Luo, Hongwei Li, Wenbo Jiang, Wenshu Fan, Zhenbo Shi, Xudong Jiang, Yi Yu,

Under Review Jan. 2026

”Evaluating Robustness of Large Audio Language Models to Audio Injection: An Empirical Study,” Conference on Empirical Methods in Natural Language Processing (EMNLP) (CCF-B, Top conference in Natrual Language Processing)

Guanyu Hou* Jiaming He*, Yinhang Zhou, Ji Guo, Yitong Qiao, Rui Zhang, Wenbo Jiang

Accepted Jul. 2025

”MTISA: Multi-Target Image-Scaling Attack,” 2024 IEEE International Conference on Communications (ICC) (CCF-C) Oral

Jiaming He, Hongwei Li, Wenbo Jiang, Yuan Zhang

Accepted Jan. 2024

Whisper in the Folds: Towards Naturalness-aware Backdoor Attacks on Protein Representation,” IEEE Transactions on Information Forensics and Security (TIFS) (CCF-A)

Jiaming He, Peijia Qin, Hongwei Li, Guanyu Hou, Shenghao Wu, Wenbo Jiang, Yi Yu, Alex C. Kot

Under Review Dec. 2025

Measuring and Harnessing Video Language Models for Automated Personal Information Extraction,” Usenix Security (CCF-A)

Jiaming He, Zhicong Huang, Guanyu Hou, Kunsheng Tang, Peigui Qi, Cheng Hong, Neil Gong

Under Review Nov. 2025

”Data Stealing Attacks against Large Language Models via Backdooring,” Electronics (JCR-Q2)

Jiaming He, Guanyu Hou, Xinyue Jia, Yangyang Chen, Wenqi Liao, Yinghang Zhou, Rang Zhou

Accepted Jul. 2024

”When Hallucinated Concepts Cross Modals: Unveiling Backdoor Vulnerability in Multi-modal In-context Learning, ACM International Conference on Multimedia in Asia (CCF-C)”

Guanyu Hou, **Jiaming He**, Yitong Qiao, Wenbo Jiang, Zihan Wang, Qiyang Song, Hongwei Li

Accepted Jan. 2026

”You Are Out of My Focus: A Defocus-Blur Backdoor Attack against Deep Learning Models, ACM International Conference on Multimedia in Asia (CCF-C)”

Tao Wang, Hongwei Li, Wenbo Jiang, **Jiaming He**, Zihan Wang, Rui Zhang, Ji Guo, Jiachen Li

Accepted Jan. 2026

”I2I Backdoor: Backdoor Attacks against Image-to-Image Networks,” IEEE Transactions on Dependable and Secure Computing (TDSC) (CCF-A)

Wenbo Jiang, Hongwei Li, **Jiaming He**, Rui Zhang, Guowen Xu, Tianwei Zhang, Rongxing Lu

Accepted Jul. 2025

”BadRefSR: Backdoor Attacks Against Reference-based Image Super Resolution,” IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP) (CCF-B)

Xue Yang, Tao Chen, Lei Guo, Wenbo Jiang, Ji Guo, Yongming Li, **Jiaming He**

Accepted Sep. 2024

”BadTTS: Identifying Vulnerabilities in Neural Text-to-Speech Models,” IEEE Global Communications Conference (Globecom) (CCF-C)

Rui Zhang, Hongwei Li, Wenbo Jiang, Ran Zhang, **Jiaming He**

Accepted Jul. 2024

SKILLS

Programming: Python, C++, Java, PyTorch, TensorFlow

Tools: $\LaTeX$, Microsoft Office Suite

Languages: English